INSTRUCTIONS

TIMED OVERCURRENT CARD

FOR USE WITH VALUTROL*

DC SCR DRIVES

TIMED OVERCURRENT CARD

193X536AAG01, G02

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company.

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*TRADEMARK OF GENERAL ELECTRIC COMPANY, U.S.A.

GENERAL

The TOC card is used for system protection by determining if the DC motor armature current exceeds rated current level for an extended time period. The G01 card initiates a drive shutdown and the G02 card provides a signal which may be used for annunciation or shutdown.

MOUNTING

The G01 card is mechanically mounted and electrically connected by screwed connections to four spade terminals marked SYS, +20V, COM and —20V to the corresponding points on the bottom row of the MF terminal in a Valutrol drive. The G02 card has three spade terminals with SYS omitted. A separate wire is run from the TOC card stab-on connector, IABS, to the corresponding connector IABS on the Interface card (IFC) for both groups of cards. The G02 card also has its output on a stab-on connector marked STO which is wired as required to perform the desired system function.

DESCRIPTION

The IABS input voltage is always negative with a nominal magnitude to 2.5V at rated drive current.

The circuit consists of an integrator (timer) stage followed by an output switching stage.

When the integrator input produced by IABS exceeds the bias voltage at IBA post the integrator output voltage at ITX starts ramping positive.

When the ITX voltage exceeds the level of the bias voltage at TTA post the output amplifier switches from a negative to a positive voltage. As the output voltage at SYS goes positive a system fault trip is initiated to shut down the drive.

In a standard drive with no modification of the bias levels, integator timing is initiated when IABS exceeds a 120% (3 volts). The nominal time to trip is—

92 seconds at 150% current, and
35 seconds at 200% current.

MODIFICATION OF CURRENT—TIME TRIP CHARACTERISTICS

In order to optimize protection or to avoid nuisance tripping it may be necessary to trim the bias level of the integrator and/or the switching stage if—

a) the IABS voltage at rated current differs from 2.5V by more than 10% or

b) the overload **rating** is other than 150% for 1 minute, or

c) the acutal system overload requirements are significantly less than the basic drive overload rating In any event, the TOC should **not** be trimmed to permit overloads in excess of those for which the equipment is designed.

The trimming is done by soldering in 1%, 1/2W resistors R3A, R4A, R10A and R11A as required to produce desired IBA and TTA bias voltages

The time to trip can be calculated from the expression —

$$T = V_{TTA} + .5 \quad 1/2\ [.29 \times |IABS| - V_{IBA}]$$

Where $|IABS|$ is the positive magnitude of the current feedback signal in volts, and T is the trip time in seconds.

From the above expression the required bias voltages can be calculated as follows:

1.a) Determine the maximum steady state magnitude $|IABS|_{ss}$ of the current feedback signal for load equal to or less than power unit, motor or A.C. supply full load operation.

b) Choose the current level, $k^1 |IABS|_{ss}$ where integration is initiated. (Allow at least 10% margin by making $k^1 = 1.1$ or larger).

c) Calculate the required integrator bias voltage $V_{IBA} = .29 \times k^1 |IABS|_{ss}$ (Volts).

d) Select R3A and/or R4A using standard resistor values to make V_{IBA} nearest the calculated value.

2. a) Determine the magnitude, $|IABS|_{OL}$, and duration in seconds, T_{OL}, of the overload.

b) Choose the current level $k^2 |IABS|_{OL}$, for which a trip should occur after T_{OL} seconds.

c) Calculate the required output bias voltage V_{TTA}

$= 1/2\ [.29 \times k^2 \times |IABS|_{OL} - V_{IBA}]\ T_{OL} - .5$ (volts).

d) Select R10A and/or R11A to make V_{TTA} closest to the calculated value.

If more than one overload rating is required, repeat the calculations for each rating and use the highest value calculated for V_{TTA}.

If the overload has a composite profile (like 250% for 10 seconds, followed by 150% for 20 seconds, etc.) calculate the average values of $IABS_{OL}$ after each load period to find the maximum required V_{TTA} voltage.

CARD TEST

Verify that the bias voltage at IBA equals +.87V ±2% (untrimmed) or is at the calculated value if trimmed.

Verify that the TTA voltage is +10V ±2% (untrimmed) or is at the calculated value if trimmed

If a Diagnostic card is furnished, and IABS reference can be applied in the DIAG STATIC mode. Otherwise, remove the wire connected to IABS and apply a test reference to IABS from an external reference potentiometer connected between —20V and common.

Apply a negative voltage to IABS and increase until the voltage at ITX starts going positive. Then reduce the IABS voltage until the ITX voltage stays fixed. The voltage at IABS should now equal 3.4 x V_{IBA}.

Increase the IABS reference until the IXT voltage ramps positive. A trip should occur (by the voltage at SYS or STO going positive) when the voltages at IXT and TTA are equal.

Reduce IABS to zero. When the voltage at IXT is reduced to about —.5V apply —5V at IABS.

Verify that the time to trip is within ±10% of the calculated value (or 35 ±3 seconds for an untrimmed card).

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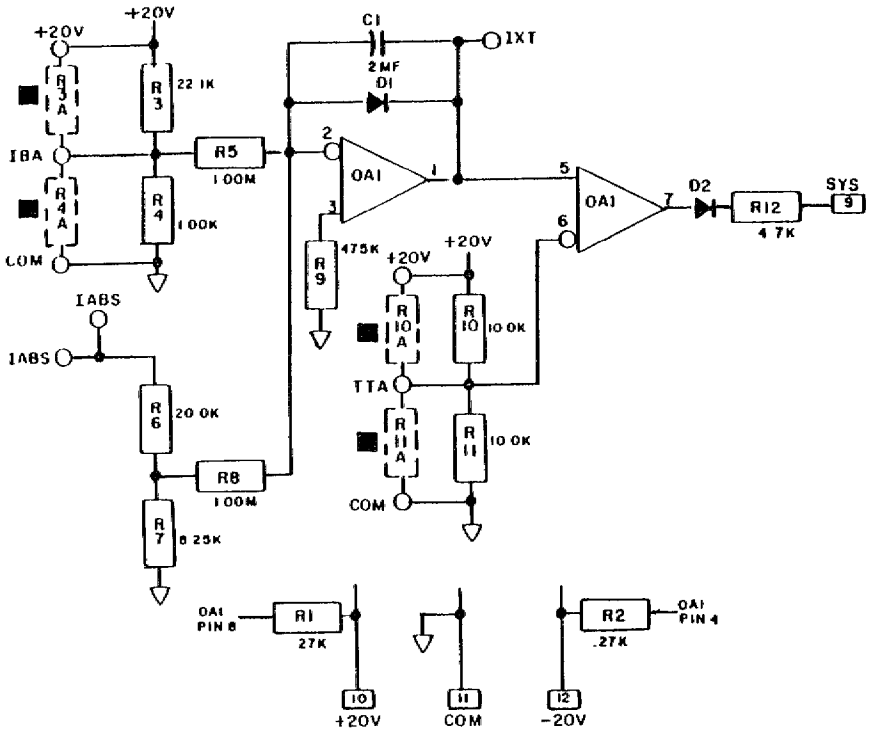
36B590233AA

CARD DIAGRAM
TIMED OVERCURRENT CARD

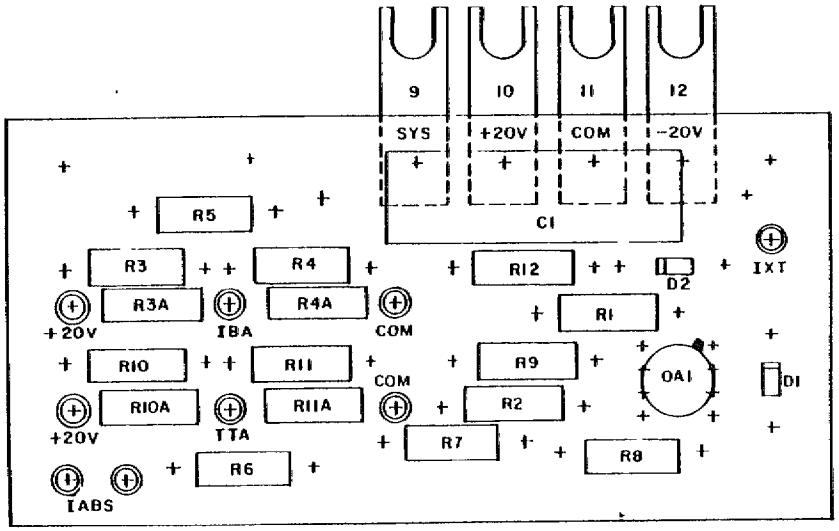
FIRST USED FOR STANDARD LINE
193X536AAG01

HOLE TABULATION

FOR PROPER HOLE DIAMETERS
SEE DRILL AND TRIM DIAGRAM
36A358175AA

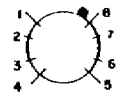


■ — OPTIONAL RESISTORS FOR MODIFICATION AS REQUIRED



NOTE
1 CARD SIZE 3 500 $^{+.000}_{-.015}$ X 1 750 $^{+.002}_{-.008}$

(TOP VIEW)



SKETCH FOR
OA1

2X SCALE	36A358031AC004 REAR ETCHED BOARD DWG

5B(8)M	5J(BW)
5R(2)BW	5D(BK)
5L(2)BW	5E(ABS)
AW(BW)	5TS(SO)
5D(CD)	
JA(CD)	

M D PAYNA MAY 18, 1976		SPEED VARIATOR	36B590233AA
P NOVAK JUNE 25, 1976		ERIE, PA	

GENERAL ELECTRIC | 36B590272AA

TITLE CARD DIAGRAM

TIMED OVERCURRENT CARD

FIRST MADE FOR STANDARD LINE

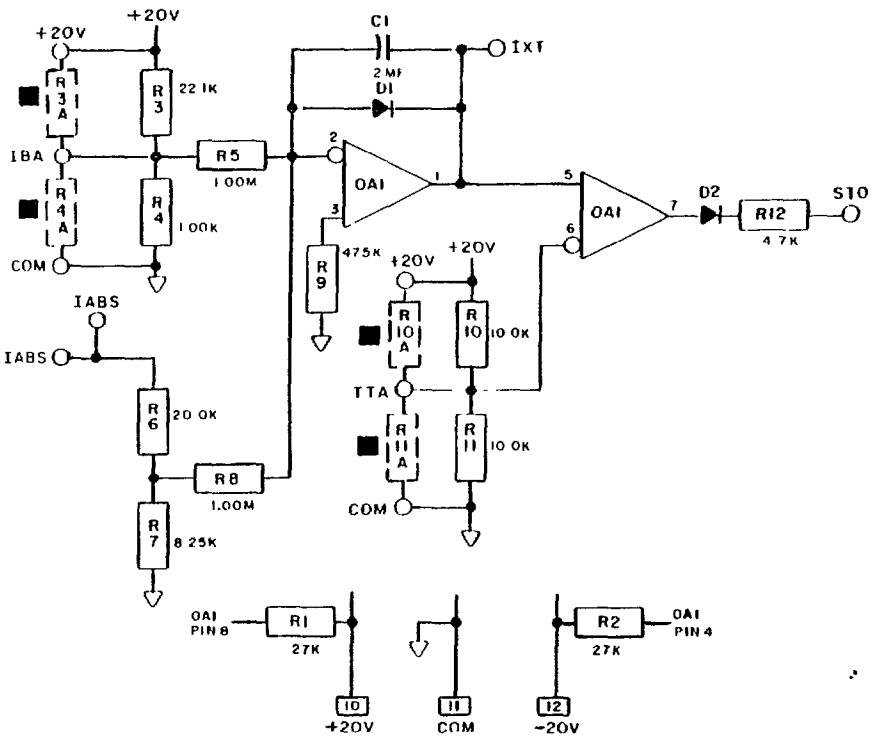
193X536AAG02

UNLESS OTHERWISE SPECIFIED PER THE FOLLOWING:

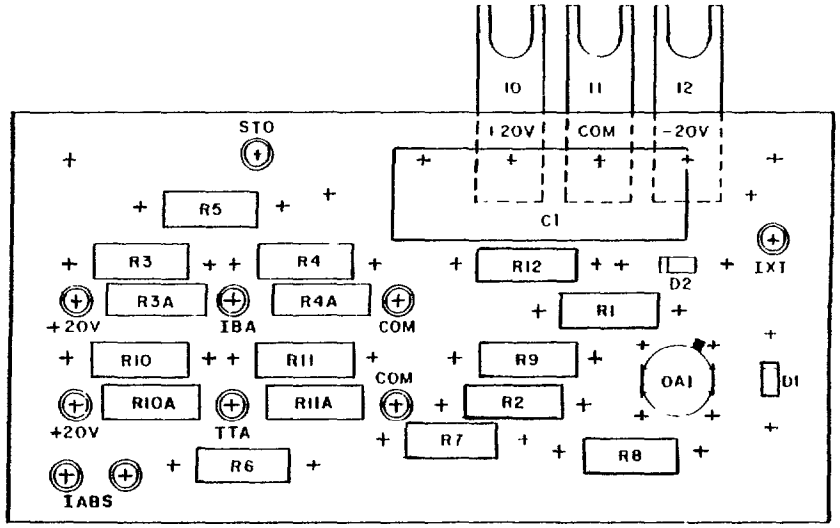
ALTERED FRACTIONS	SURFACES				0
	✓				

HOLE TABULATION

FOR PROPER HOLE DIAMETERS
SEE DRILL AND TRIM DIAGRAM
36A358175AA

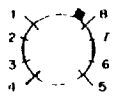


■ — OPTIONAL RESISTORS FOR MODIFICATION AS REQUIRED



NOTE:
1 CARD SIZE $3.500^{+.000}_{-.015}$ X $1.750^{+.002}_{-.008}$

(TOP VIEW)



SKETCH FOR OAI

2X	36A358031AC004
SCALE	REAR ETCHED BOARD DWG

REVISIONS	PRINTS TO	
	5B(8)M	5J(BW)
	5R(2)BW	5D(BK)
	5L(2)BW	6E(ABS)
	AW(BW)	6TS(50)
	5D(CD)	
	JA(CD)	

NO REVISIONS ARE TO BE MADE TO THIS DRAWING WITHOUT THE SPECIFIC APPROVAL OF THE DEVELOPMENT ENGINEERING SECTION OF THE SPEED VARIATOR DEPARTMENT

DRAWN BY (A. Sykes) OCT 23,1981	(11)	Speed Variator ERIE, PA, USA	36B590272AA
CHECKED (C. Hyde) JULY 9, 1982			

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GENERAL ELECTRIC COMPANY
SPEED VARIATOR PRODUCTS OPERATION
ERIE, PENNSYLVANIA 16531

GENERAL ELECTRIC

GEK-45119B (10/85) 1M (P)